

Fast laser camera that can see around corners

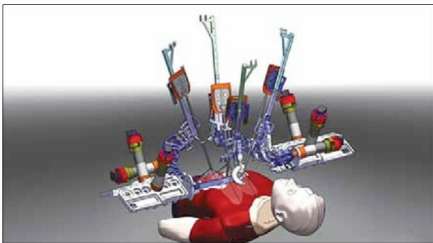
Ever wondered why you can hear but not see around corners? Now a camera developed by the Camera Culture group at MIT can see around corners to capture 3D images of objects that would otherwise be hidden from sight. A potential application of the camera is to spot objects in hazardous areas.

The crux of the technique is the ability of the camera to form images from light waves that reflect off the walls in every direction. To do so, the camera fires very short laser pulses at a flat wall that scatters light onto the hidden object. The object then scatters some of that light back toward the wall. For two trillionths of a second, the high-speed camera collects light hitting the wall. Changing the interval between the laser flash and the camera snap measures light returned at different times, corresponding to parts of the object at different distances.

The resolution is limited by the speed of light, which travels about 0.6 millimetre during the two trillionths of a second of each camera frame, and by the fact that light scattered on different paths can return to the camera at the same time.

Gesture control for unmanned aircraft

Gesture recognition is finding its way into unmanned aircraft control. Researchers at Massachusetts Institute of Technology have devised a gesture recognition system using which flight crew can command a drone aircraft. Drones are unmanned aerial vehicles that can land autonomously but are controlled when taxi-



Linux-based surgical robot

ing on a flight deck.

Developed by Yale Song, Randall Davis and David Demirdjian, the system can discern and recognise gestures that command the aircraft to perform functions such as turning off the engine or travelling to a refuelling point. The system is somewhat similar to Microsoft's Kinect. Clearly defined gestures are picked up by a depth-sensing camera. The image sensor captures and analyses 24 body and hand motions pretty fast with 76 per cent degree of accuracy.

The team is currently working towards improving recognition levels. The study could ultimately pave the way for pilots and flight crews to direct unmanned planes on the flight deck in a real-time aircraft carrier environment.

Linux-based surgical robot

Researchers at the University of Washington, in Seattle, are about to release numerous robotic surgeons called 'Raven.' Raven is the first surgical robot to use Open Source software. With the help of Ravens, surgeons can make tiny incisions that cause less tissue damage,

enabling the patient to recover fast.

The Linux-based operating system of the robot allows anyone to modify and improve the original code. The robots have wing-like arms, are lightweight and compact, and cost around \$250,000.

American regulators have not approved Ravens yet and present experiments are restricted to only animals and human cadavers. Most robot-assisted surgery today is performed using the da Vinci surgical system. Unlike Raven, da Vinci system is immobile, weighs more than half a tonne, and costs \$1.8 million. It uses proprietary software, which cannot be modified.

Dr Hannaford is looking for help from robotics researchers and amateurs to find and fix bugs in the Open Source code of Raven.

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